MIL-STD-399A
4 December 1976

MILITARY STANDARD

MICROFORM FORMATS

FSC MISC

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DEPARTMENT OF DEFENSE
WASHINGTON D.C. 20301

Microform Formats

MIL-STD-399

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, SARPA-TS-R, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

The use of microforms in the recording and dissemination of data and information is widespread in the Department of Defense, as well as outside of the DoD, and the volume is steadily increasing. Uniformity of microform formats is therefore essential in order to enhance exchange and utilization of recorded information, and to provide compatibility of processing and user equipment. It is also essential that the number of different microform formats be kept to a minimum in order to reduce equipment proliferation. This standard has been developed and promulgated to meet these needs.

Usage of this standard has revealed some incompatibilities between those requirements prescribed in the standard, and related requirements prescribed in those military standards and specifications from which the formats were extracted when this standard was initially developed. The purpose of this revision is to reduce these incompatibilities.

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1. SCOPE

1.1 Scope. This standard establishes formats for microforms produced by or for the Department of Defense.

1.2 Purpose. The purpose of this standard is to (1) assure uniformity of dimensions, image placement and orientation for microforms produced by or for the Department of Defense, which will enhance the exchange and utilization of data and information between and among Department of Defense components; (2) provide for compatibility of processing and user equipment, (filming equipment, viewers, etc.) and (3) preclude the unnecessary proliferation of types and varieties of processing and viewing equipment by controlling and minimizing microform formats.

1.3 Application. This standard applies to microforms produced by or for any DoD activity.

1.3.1 Specified microform formats. The microform formats specified in 5.4 through 5.11 (Formats 1 through 8) are the only formats of these microforms approved for production by or for the Department of Defense and are mandatory for all new microform applications.

1.3.2 Existing microform systems. Existing microform systems are not required to convert. Nevertheless, systems not in accordance with this standard shall be evaluated periodically and the merits of converting to the standard weighed.

1.3.3 Choice of formats. The choice of microform format or formats from among those authorized herein for use in a particular user application is not prescribed by this standard. These choices will be based on user needs and developed from other criteria including handbooks, manuals and specific studies of the subject by the user. However, where microfiche is chosen as the most suitable microform, format "5" is recommended for non-COM applications and format "6" is recommended for COM applications.

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1.4 Classification of microform formats. Following are the authorized microform formats (See Paragraphs 5.4 thru 5.11):

Format 1 - 16mm Roll Microfilm - Paragraph 5.4
Format 2 - 35mm Roll Microfilm - Paragraph 5.5
Format 3 - Aperture, Camera, and Copy Cards - Paragraph 5.6
Format 4 - 63 Frame 24:1 Reduction Microfiche - Paragraph 5.7
Format 5 - 98 Frame 24:1 Reduction Microfiche - Paragraph 5.8
Format 6 - 270 Frame 48:1 Reduction Microfiche - Paragraph 5.9
Format 7 - 420 Frame 48:1 Reduction Microfiche - Paragraph 5.10
Format 8 - 105mm Filmed Reproduction - Paragraph 5.11

2. Referenced documents

2.1 The following documents of the issues in effect on the date of invitation for bids or requests for proposals form a part of this Standard to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-9868	Microfilming of Engineering Documents, 35mm, Requirements for
MIL-C-9877	Cards, Aperture
MIL-P-9879	Photographing of Construction/Architectural Drawings, Maps and Related Documents, 105mm; Requirements for
MIL-C-9949	Cards, Copy
MIL-M-38748	Microfiche: For Engineering/Technical Data, Reports, Studies and Related Data, Requirements for
MIL-M-38761	Microfilming and Photographing of Engineering/Technical Data and Related Documents: PCAM Card Preparation, Engineering Data Micro-Reproduction System, General Requirements for Preparation of
MIL-M-46849	Microfilming of Engineering and Related Documents, 16mm, Requirements for

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MIL-F-80242 Film Microfiche, 48X

STANDARDS

Military

MIL-STD-804 Formats and Coding of Aperture, Copy and Tabulating Cards for Engineering Data Micro-Reproduction System

Industry

ASTM-E-380-76 Metric Practice Guide

PUBLICATIONS

Directive

Department of Defense DoD Directive 5200.1-R Information Security Program Regulation

Manual

Defense Standardization Manual 4120.3M, Standardization Policies, Procedures and Instructions

(Copies of specifications, standards, handbooks, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

(Copies of these government documents may be obtained by authorized government offices by completing DD Form 1425 and processing that form as indicated thereon.)

2.2 Other publications. The following documents form a part of this standard to the extent specified herein (see 4.3.2)

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STANDARDS

American National Standards Institute (ANSI)

PH5.3-1967 (R.1973)	16mm and 35mm Silver-Gelatin Microfilms for Reel Applications, Specifications for
PH5.8-1972	Unitized Microfilm Carriers (Aperture, Camera, Copy and Image Cards), Dimensions for
PH5.9-1975	Microfiche of Documents
PH5.12-1973	Measuring Thickness of Buildup Area on Unitized Microfilm Carriers (Aperture, Camera, Copy and Image Cards), Method for
PH5.14-1973	Determining Adhesion of Protection Sheet to Aperture Adhesive of Unitized Microfilm Carrier (Aperture Card), Method for
PH5.18-1976	Format and Coding Standards for Computer Output Microfilm
PH5.20-1974	Document Mark (Blip) used in Image Mark Retrieval Systems

(ANSI standards may be obtained from the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018). See Note.

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Note - Specifications and standards of technical societies and technical associations are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies. DoD adopted industrial standards may also be obtained by DoD activities from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

3. Definitions

3.1 Definitions needed for clarification of this Standard are given in this section. These definitions do not supersede or abridge those used by the procuring agency.

3.1.1 Aperture card. A tabulating card with a rectangular hole specifically prepared for the mounting of a frame of microfilm (Source: Specification MIL-M-38761).

3.1.2 Cartridge. A single core container enclosing roll microfilm, designed to be inserted into viewers, viewer/printers and retrieval and camera devices.

3.1.3 Cassette. A double core container enclosing processed roll microfilm designed to be inserted into viewers, viewer/printers and retrieval devices.

3.1.4 Column. A vertical series of images on a microform.

3.1.5 COM. (1) Computer Output Microfilm: microfilm containing data produced by a recorder from machine generated electrical signals.

(2) Computer Output Microfilmer: a recorder which converts data from a machine into human intelligible form and records it on microfilm.

(3) Computer Output Microfilming: a method of converting data from a machine into human intelligible form onto microfilm.

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3.1.6 Copy card. A tabulating card with a frame of unexposed and unprocessed microfilm mounted in or over a rectangular hole for subsequent exposure and development of the microfilm which is mounted in the card. (Source: Specification MIL-M-38761).

3.I.7 Cutmark. A mark added to film to permit automatic cutting of microfiche from a roll of film, or use of automatic machinery designed to utilize a cutmark.

3.1.8 Film. Any sheet or strip of transparent plastic coated with a light-sensitive layer.

3.1.9 Format, microform. A dimensioned layout containing requirements for size, placement and orientation of micro-images upon a microform.

3.1.10 Frame, microfilm. The total area of microfilm utilized in one exposure, regardless of whether or not this area is filled by the document image.

3.1.11 Frame, microfiche. (1) One microimage and margin contained within a film frame. (2) A single area of a grid pattern.

3.1.12 Heading. Inscription placed at top of microfiche to identify its contents. Also referred to as Title, it is readable without magnification.

3.1.13 Image. A representation of an object such as a document or other information sources produced by light rays.

3.1.14 Image orientation. The arrangement of objects or images with respect to the edges of the film.

3.1.15 Jacket. A transparent plastic carrier with single or multiple sleeve or pocket made to hold microfilm in flat strips.

3.1.16 Microfiche. A sheet of film containing multiple microimages in a grid pattern. It usually contains a heading or title which can be read without magnification.

3.1.17 Microfilm. (1) A fine grain, high resolution film containing an image or images greatly reduced in size from the original.

(2) The recording of microimages on film
(3) Raw film with characteristics as in (1)

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3.1.18 Microform. A generic term for any form, whether film, video tape, paper or other medium, containing miniaturized or otherwise compressed images which cannot be read without special display devices.

3.1.19 Micrographics. The science, art and technology of document and information miniaturization and associated microform systems.

3.1.20 Military "D". The term given to the DoD specified aperture size and location in (EAM) tabulating cards.

3.1.21 Reader - See viewer.

3.1.22 Reader-Printer - See viewer/printer.

3.1.23 Reduction. A measure of the number of times a given linear dimension of an object is reduced when photographed, expressed as 24:1, 48:1, etc.

3.1.24 Reduction Effective. A measure of the number of times a given linear dimension of a similar conventional document would be reduced to equal the size of the COM generated micro-image.

3.1.25 Row. A horizontal series of microimages on microform.

3.1.26 Specification. A document intended primarily for use in procurement, which describes the essential and technical requirements for items, materials, and services including procedures by which it will be determined that the requirements have been met. (Source: Defense Standardization Manual 4120.3-M).

3.1.27 Standard. A document that establishes engineering and technical limitations and applications for items, materials, processes, methods, designs and engineering practices. (Source: Defense Standardization Manual 4120.3-M).

3.1.28 Tabulating Card. A card on which data are entered by use of punched holes or other means that can be sensed by a machine so that it can sort, collate, list, total, or otherwise manipulate the card or the data.

3.1.28 Viewer. A device for viewing and reading a microform with the unaided eye.

3.1.30 Viewer/Printer. A machine which combines the functions of a viewer and an enlarger-printer.

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4 GENERAL REQUIREMENTS

4.1 Dimensioning

4.1.1 The dimensions used in Formats 1 through 8 are expressed in the metric system, all dimensions shown being in millimeters. Conversion tables for converting from millimeters to inches are contained in Standard for Metric Practice (IEEE-268-1975, ANSI Z210.1) ASTM-E 380-76 and are available from other sources as well.

4.1.2 The width dimensions of the roll microfilm shown on Formats 1 and 2 are of the raw stock immediately after cutting (See ANSI PH5.3-1967 (R. 1973)). The external dimensions of microfiche shown on Formats 4, 5, 6 and 7 apply immediately after processing (See PH5.9-1975).

4.2 Requests for Deviations. Requests for deviations from this standards should be forwarded to the office of primary interest for microforms management in the department for further referral to the Office of the Assistant Secretary of Defense (Installations and Logistics), Defense Materiel Specifications and Standards Office, Cameron Station, Alexandria, VA 21314. Approval of deviations will be limited normally to microform systems exclusively within a department or agency.

The offices of primary interest for microforms management and their authority are as follows:

	AUTHORITY
NAVY Chief of Naval Operations OPNAV (Op-09B17) Washington, D. C. 20350	
MARINE CORPS Headquarters, Marine Corps HQMC (ISM) Washington, D. C. 20380	
AIR FORCE HQ USAF (AF/DAYX) Washington, D. C. 20330	
ARMY HQDA (DAAG-AMS-M) Washington, D. C. 20314	AR 340-22: The Army Micrographics Program
DSA Director, Defense Supply Agency ATTN: DSAH-SE Cameron Station Alexandria, VA 22314	

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4.3 Utilization of industry standards

4.3.1 Policies, procedures and instructions for utilization by the Department of Defense of Industry standardization documents are set forth in Defense Standardization Manual 4120.3-M. Basically, in instances where industry standardization documents are available that satisfy the needs of the Military, these documents will be used in lieu of preparing Military or Federal documents. In view of the fact that economical and ready dissemination and exchange of data and information are the main purpose for the use of microforms, and in view of the extensive and increasing use of microforms in Industry (as well as in the DoD) it is incumbent that full advantage be taken of Industry standardization documents in this area. Accordingly, Industry standards are utilized wherever relevant in this standard and to the extent specified.

4.3.2 Revision of industry standards. Revisions made in Industry standards may affect the application of these standards in the Department of Defense. Therefore, the Industry documents referenced herein and bearing the issue dates shown are the only issues which apply, except where specifically otherwise approved by the Procuring Agency. They apply only to the extent shown.

4.4 Microfiche index. An index to a microfiche is not required but is encouraged to permit ready access to information contained within the microfiche. Self indexing systems, such as personnel applications, do not need additional indexing. If an index to the microfiche is to be provided, the placement of the last micro image of the index shall be in the bottom right hand corner frame of the grid area. Additional index images will be placed in the preceding frames as required.

5. DETAILED REQUIREMENTS

5.1 Order of precedence of standardization documents. The following order of precedence applies for requirements shown in this standard and the standardization documents listed and referenced herein. Whenever conflicting requirements exist, precedence shall be determined in the same order.

5.1.1 First precedence. Requirements for microform formats and reduction ratios prescribed in this standard.

5.1.2 Second precedence. Requirements prescribed in Military and Federal documents listed in 2.1.

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5.1.3 Third precedence. Requirements prescribed in Industry Documents listed in 2.2.

Note: Additional Order of Precedence information can be obtained from MIL-STD-143, Standards and Specifications, Order of Precedence for the Selection of (see Appendix).

5.2 Reduction and sectionalizing.

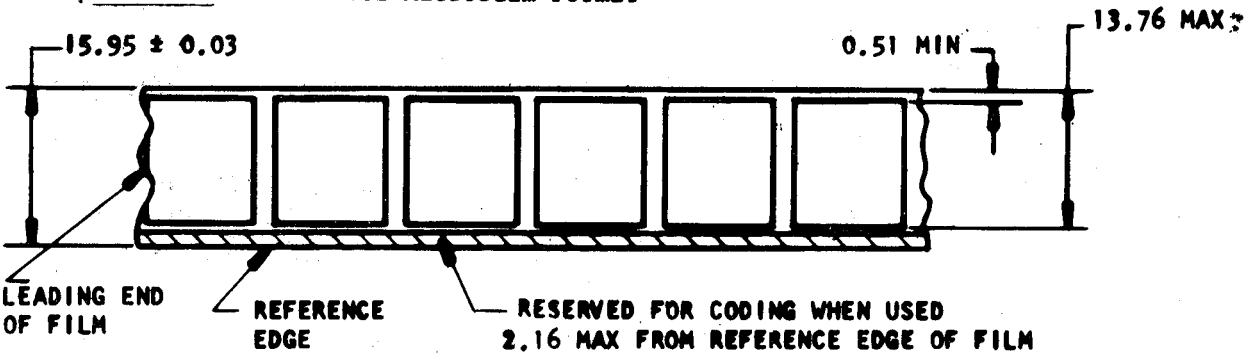
5.2.1 Reduction (effective reduction) ratios of microfiche are 24:1 and 48:1. The criteria for selecting the appropriate reduction ratio is the microfiche format used. See formats 4 through 7 of this standard for microfiche formats. Reductions used in other microform formats covered by this standard are prescribed in the applicable microform specifications and standards referenced below the formats.

5.2.2 Documents which are too large to fit in one frame must be sectionalized. Details as to procedures, including amount of overlap, are given in documents referenced herein. Sectionalizing is also required where the quality of the document, small lettering size, etc. so dictate.

5.3 Classified microforms. Microforms containing classified information shall be treated as to marking, handling and downgrading in accordance with DoD Directive 5200.1-R and detailed DoD instruction, regulations and procedures derived therefrom including those given in the standardization documents referenced herein.

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5.4 Format 1. 16mm Roll Microfilm Format



Dimensions shown are those applying in the plane shown, and are required to assure uniformity in image, size, placement and orientation, and compatibility of processing and user equipment. This format applies to both COM and Non-COM generated 16mm roll microfilm. Additional specifications for reduction, film thickness, frame and image sizes, orientation, coding and other details (except that all requirements shown on Format 1 shall take precedence) are contained in the Industry and Military specifications/standards shown below.

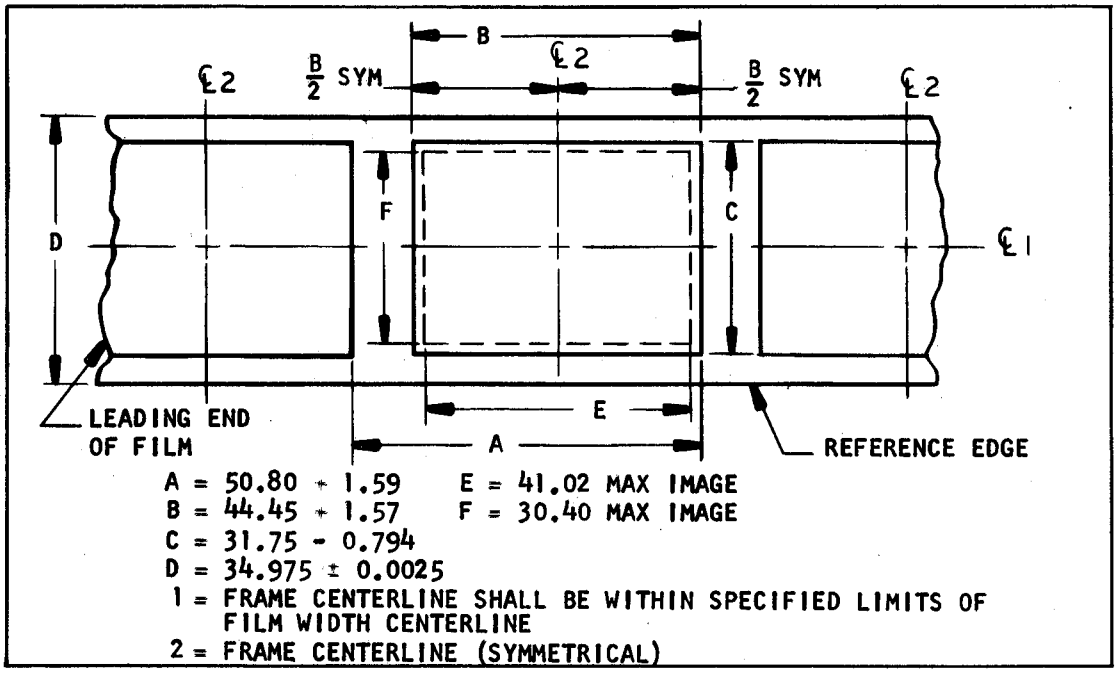
RELATED STANDARDS/SPECIFICATIONS

Industry Standards	Military Specification/Standards
ANSI PH5.18-1976, Format and Coding Standards for Computer Output Microfilm, Section 5.2	MIL-M-81930(AS), Mil Spec, Manuals, Technical: 16mm Silver Halide Microfilm, Requirements for (See Note).
ANSI PH5.20-1974, Document Mark (Blip) used in Image Mark Retrieval Systems	MIL-M-81931 (AS), Mil Spec. Manuals, Technical: 16mm Duplicate Microfilm Cartridge Loading and Labeling, Packaging and Shipping; Requirements for (See Note).
ANSI PH5.3-1967 (R.1973), 16mm and 35mm Silver-Gelatin Microfilms for Reel Applications, Specifications for	MIL-M-46849, Microfilming of Engineering and Related Documents, 16mm Requirements for.
	MIL-M-38802 (USAF), Microfilming, 16mm and Reproduction of Technical Orders, Requirements for (See Note)

NOTES - These specifications are listed in the Appendix and are not requirements of this standard.

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5.5 Format 2. 35mm Roll Microfilm Format.



Dimensions shown are those applying in the plane shown, and are required to assure uniformity in image, size, placement and orientation and compatibility of processing and user equipment. This format applies for both COM and Non-COM-Generated 35mm Roll Microfilm. Specifications for reduction, additional dimensions (thickness, etc.), orientation, coding, and other details (except that all requirements shown on Format 2 shall take precedence) are contained in the Industry and Military specifications/standards shown below.

RELATED STANDARDS/SPECIFICATIONS

Industry Standards

ANSI PH5.18-1976, Format and Coding Standards for Computer Output Microfilm, Section 4

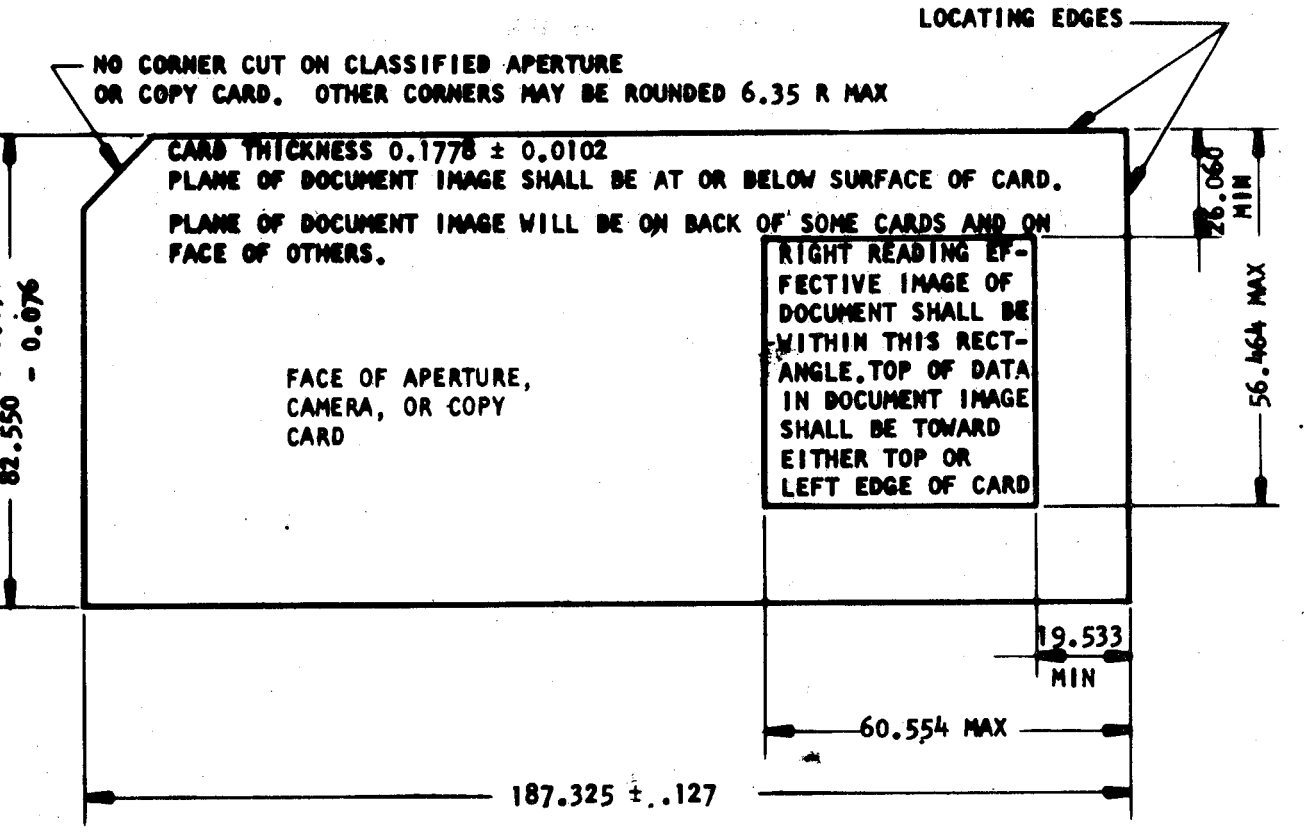
ANSI PH5.3-1967 (R.1973), 16mm and 35mm Silver-Gelatin Microfilms for Reel Application, Specifications for

Military Specifications/Standards

MIL-M-9868, Microfilm of Engineering Documents, 35mm, Requirements for

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5.6 Format 3. Aperture, Camera, and Copy Card Size and Document Image Location



Specifications for materials, quality, coding, copying, and other details (except that all requirements shown on Format 3 shall take precedence) are contained in the Industry and Military specifications/ standards shown below.

RELATED STANDARDS/SPECIFICATIONS

Industry Standards

ANSI PH5.8-1973, Dimensions for Unitized Microfilm Carriers (Aperture, Camera, Copy and Image Cards) - "D" Aperture

ANSI PH5.12-1972, Measuring Thickness of Build-up Area on Unitized Microfilm Carriers (Aperture, Camera, Copy and Image Cards), Method for
ANSI PH5.14-1973, Determining dhesion of Protection Sheet to Aperture Adhesive of nitized Microfilm Carrier (erture Card), Method for

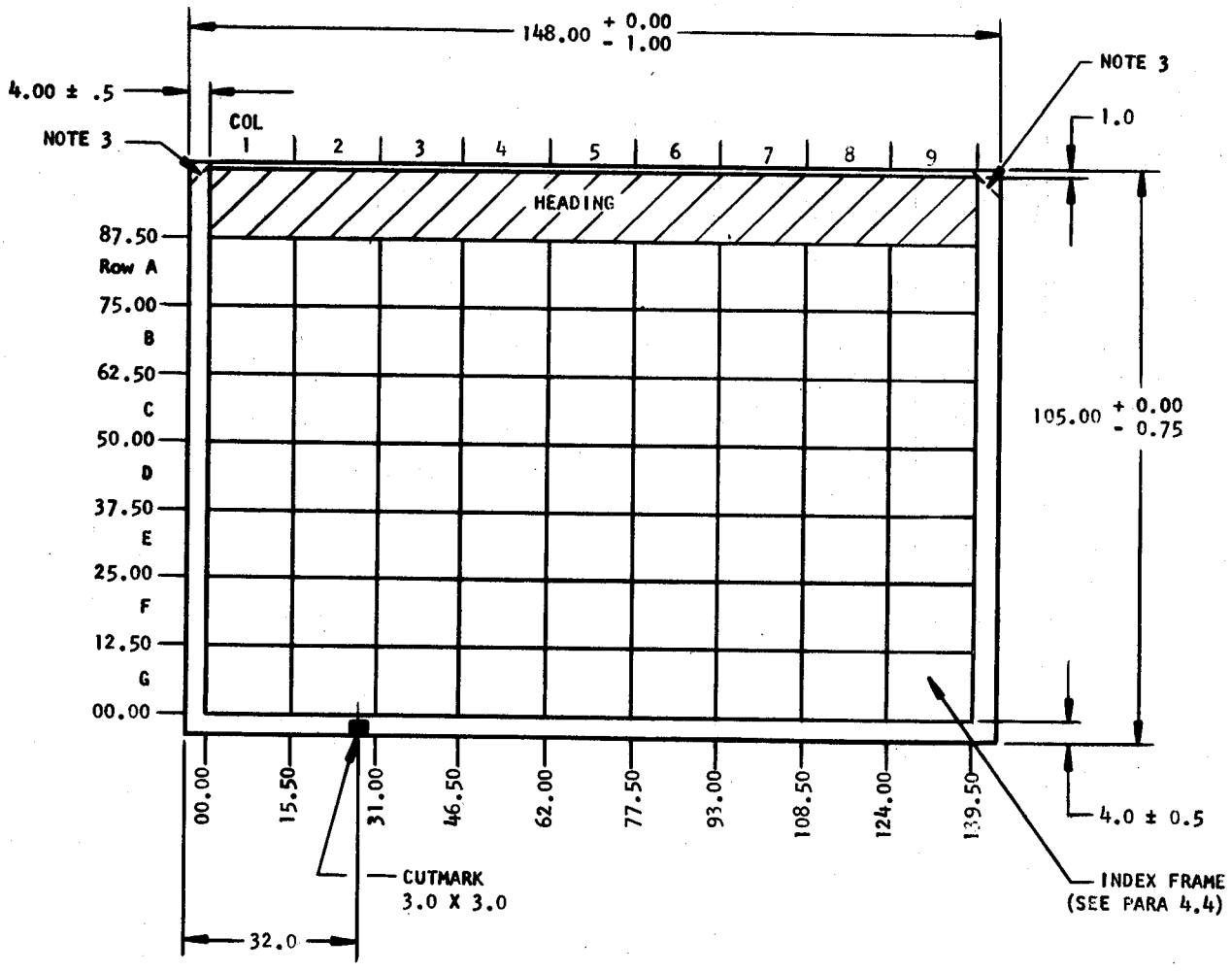
Military Specifications/Standards

MIL-STD-804, Formats and Coding of Aperture, Camera, Copy and Tabulating Cards for Engineering Data Micro-Reproduction System

MIL-C-9877, Cards; Aperture, Camera, and Copy
MIL-C-9949, Cards, Copy.
MIL-M-38761, Microfilming and Photographing of Engineering/Technical Data and Related Documents: PCAM Card Preparation, Engineering Data Micro-Reproduction Systems, General Requirements for, Preparation for

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5.7 Format 4. 63 Frame 24:1 Reduction Microfiche Format (Note 4)



Notes:
1. Grid lines shown are for dimensional representation and are not required to appear on finished microfiche.
2. Additional rows starting with Row A may also be used for heading.
3. Corner cut as specified in procurement specification/contract.
4. Designed for 14" x 11" format documents.

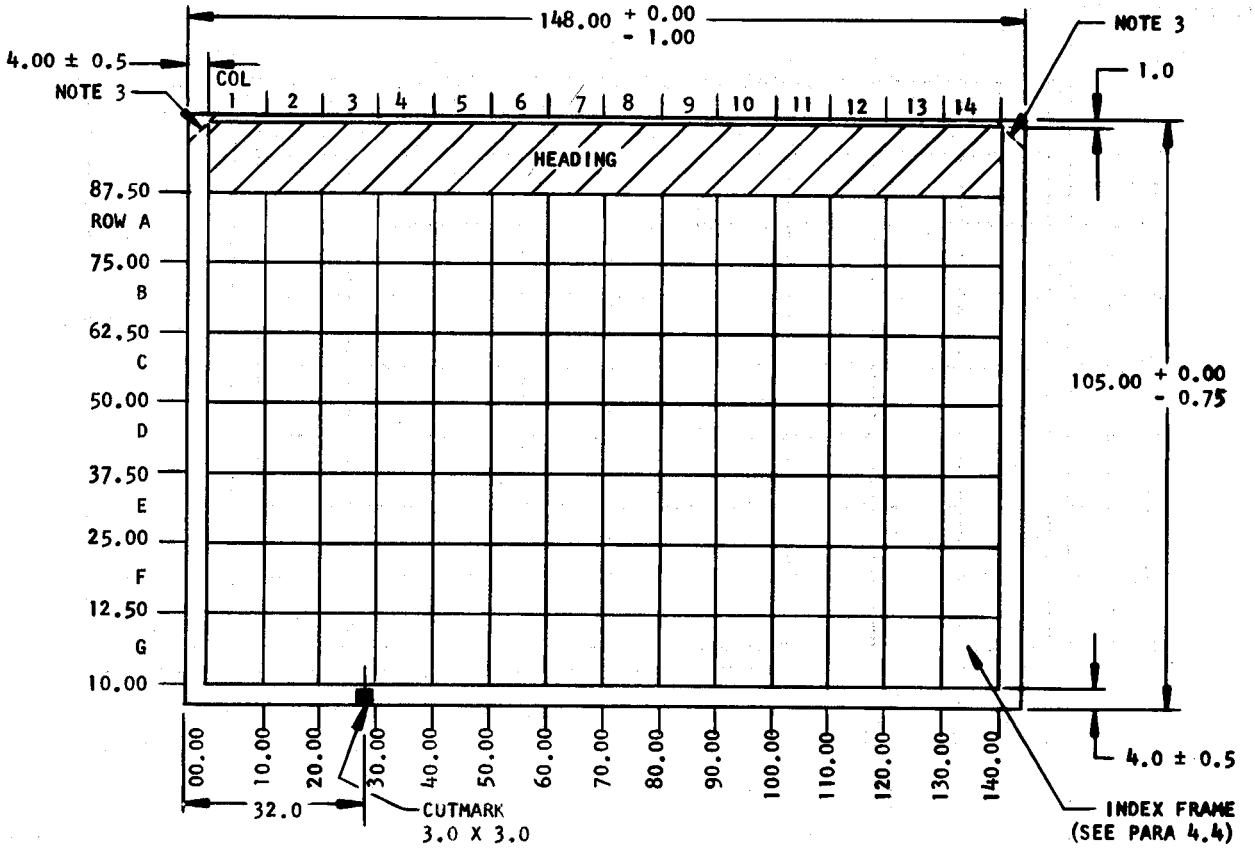
Dimensions shown are those applying in the plane shown, and are required to assure uniformity in image, size, placement and orientation, and compatibility of processing and user equipment. This format applies for both COM and Non-COM-Generated 63 Frame Microfiche. Specifications for additional dimensions (thickness, etc.), orientation, coding and other details (except that all requirements shown on Format 4 shall take precedence) are contained in the Industry and Military specification/standards shown below.

RELATED STANDARDS/SPECIFICATIONS

Industry Standards	Military Specifications/Standards
ANSI PH5.18-1976, Format and Coding Standards for Computer Output Microfilm, Section 5.3, Format A3	None

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5.8. Format 5. 98 Frame 24:1 Reduction Microfiche Format



Notes:
1. Grid lines shown are for dimensional representation and are not required to appear on finished microfiche.
2. Additional rows starting with Row A may also be used for heading.
3. Corner cut as specified in procurement specification/contract.
4. Designed for 8 1/2" x 11" format documents.

Dimensions shown are those applying in the plane shown, and are required to assure uniformity in image, size, placement and orientation, and compatibility of processing and user equipment. This format applies for both COM and Non-COM-Generated 98 Frame Microfiche. Specifications for additional dimensions (thickness, etc.), orientation, coding, and other details (except that all requirements shown on Format 5 shall take precedence) are contained in the Industry and Military specifications/standards shown below.

RELATED STANDARDS/SPECIFICATIONS

Industry Standards

ANSI PH5.18-1976, Format and Coding Standards for Computer Output Microfilm, Section 5.3, Format A1

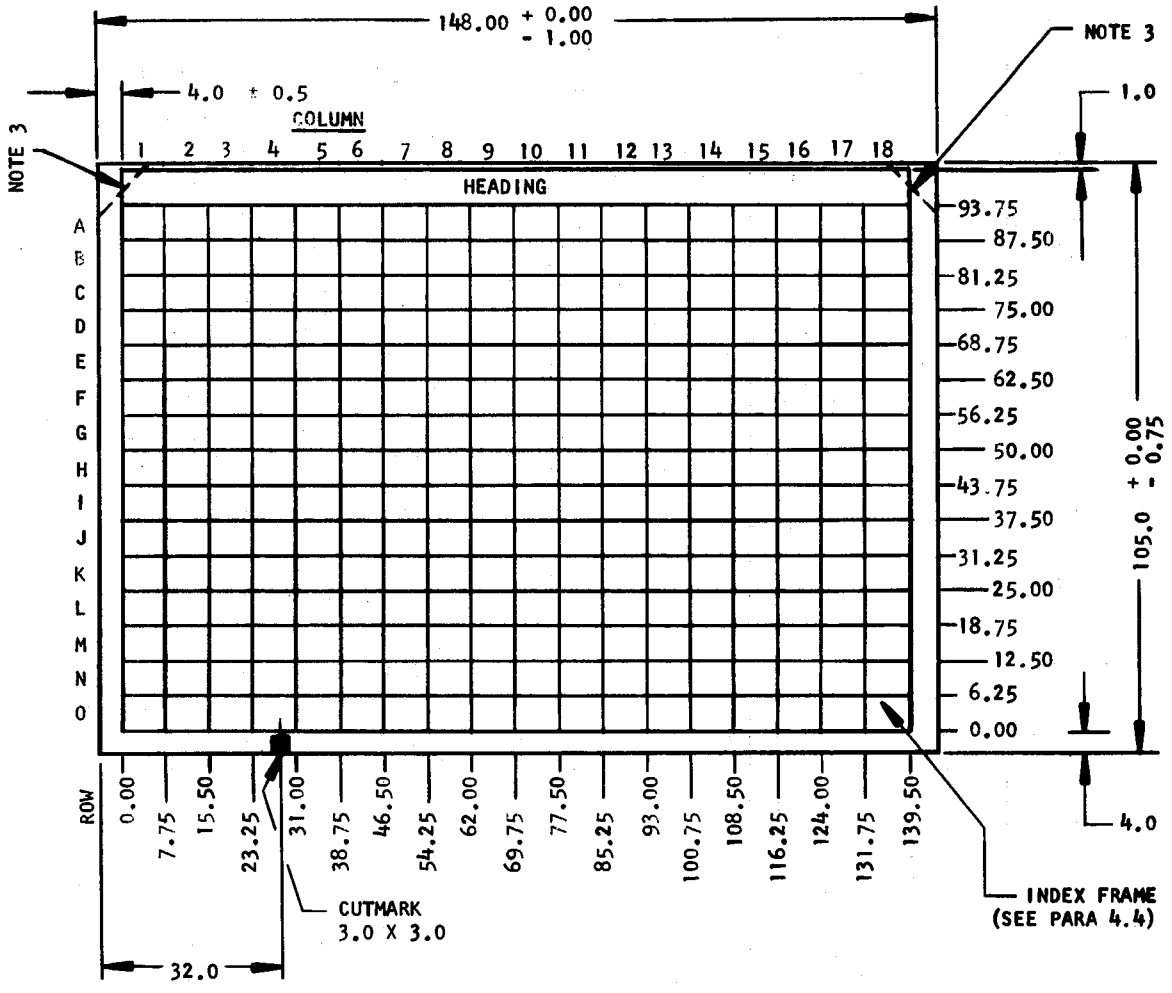
ANSI PH5.9-1975, Microfiche of Documents

Military Specifications/Standards

MIL-M-38748 Microfiche; For Engineering/ Technical Data, Reports, Studies and Related Data, Requirements for.

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5.9 Format 6. 270 Frame 48:1 Reduction Microfiche Format



Notes:
1. Grid lines shown are for dimensional representation and are not required to appear on finished microfiche.
2. Additional rows starting with Row A may also be used for heading.
3. Corner cut as specified in procurement specifications/contact.
4. Designed for 14" x 11" format documents.

Dimensions shown are those applying in the plane shown, and are required to assure uniformity in image, size, placement and orientation, and compatibility of processing user equipment. This format applies for both COM and Non-COM-Generated 270 Frame Microfiche. Specifications for additional dimensions (thickness, etc.), orientation, coding, and other details (except that all requirements shown on Format 6 shall take precedence) are contained in the Industry and Military specifications/standards shown below.

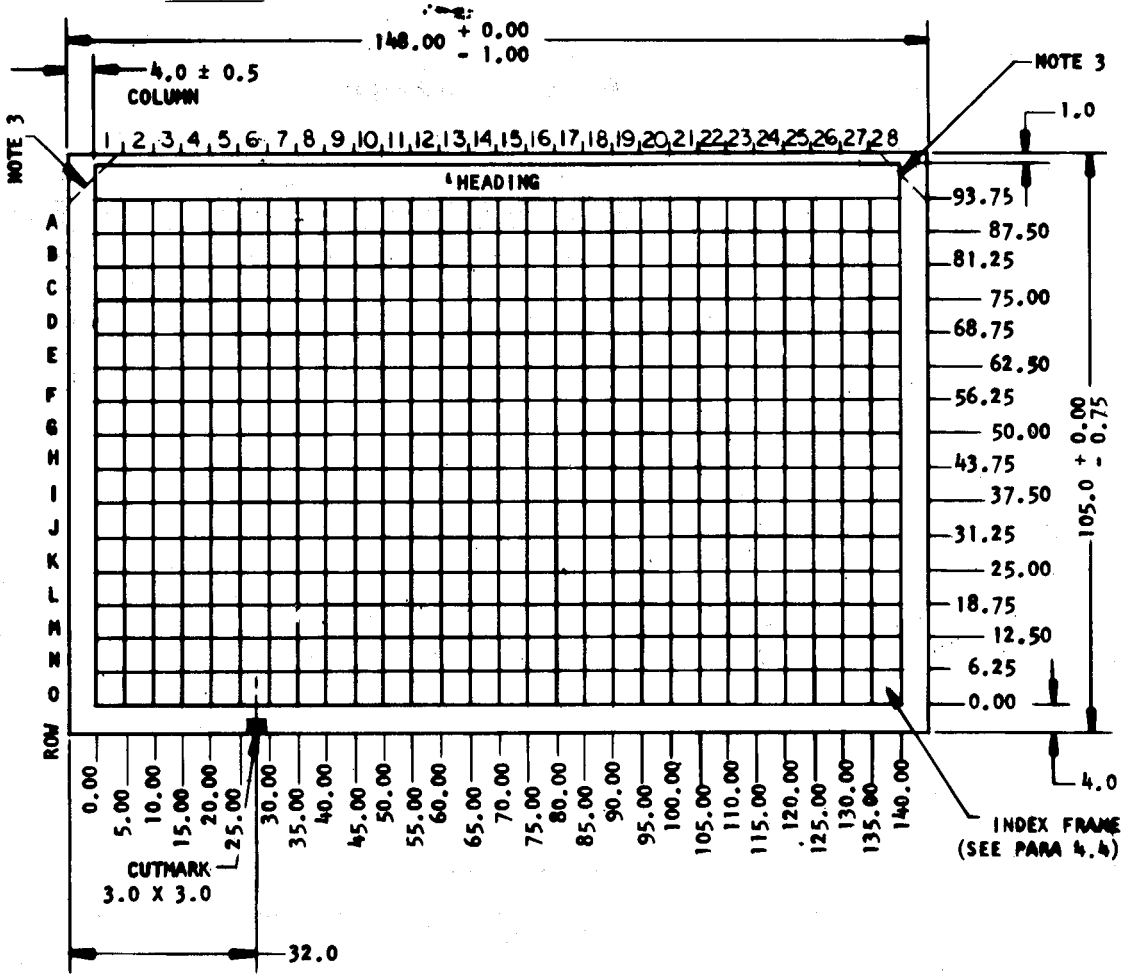
RELATED STANDARDS/SPECIFICATIONS

Industry Standards
Presently none

Military Specifications/Standards
MIL-F-80242, Military Specification, Film, Microfiche 48X

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5.10 Format 7. 420 Frame 48:1 Reduction Microfiche Format.



Notes:
1. Grid lines shown are for dimensional representation and are not required to appear on finished Microfiche.
2. Additional rows starting with Row A may also be used for heading.
3. Corner cut as specified in Procurement Specification/Contract.
4. Designed for 8.5" x 11" format documents.

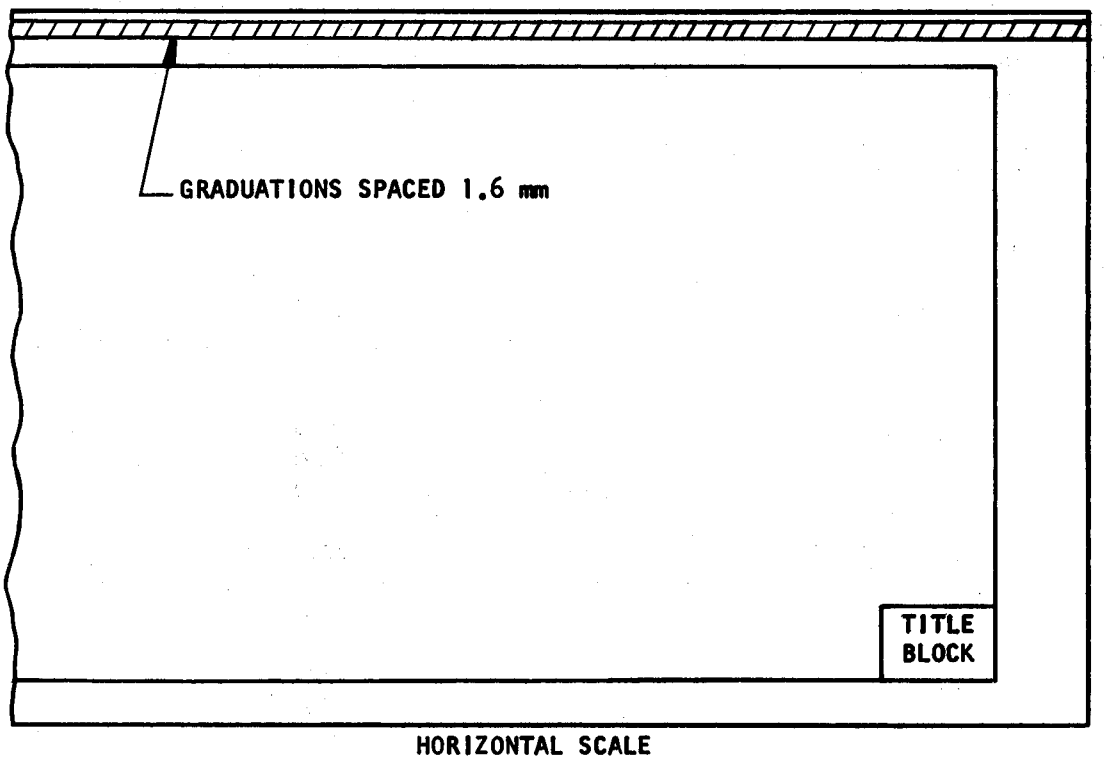
Dimensions shown are those applying in the plane shown, and are required to assure uniformity in image, size, placement and orientation, and compatibility of processing and user equipment. This format applies for both COM and NonCOM-Generated 420 Frame Microfiche. Specifications for additional dimensions (thickness, etc.), orientation, coding, and other details (except that all requirements shown on Format 7 shall take precedence) are contained in the Industry and Military specification/standards shown below.

RELATED STANDARDS/SPECIFICATIONS

Industry Standards	Military Specifications/Standards
None	None

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5.11 Format 8. 105mm Filmed Reproduction Format.



This format applies for 105mm Filmed Reproductions of Construction/ Architectural Drawings, Maps and Related Documents.

Specifications for reduction, size, quality, resolution, image location, and other details are contained in the following Military specification:

MIL-P-9379, Photographing of Construction/Architectural Drawings, Maps, and Related Documents, 105mm; Requirements for

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6. APPENDIX
(Supplementary list of Microform Documents for information and use)

6.1 Scope of appendix. The scope of this appendix is to identify microform documents that are not requirements of this standard but are referenced for information and appropriate use.

6.2 Governmental documents.

SPECIFICATIONS

Military

MIL-M-9868/1(ships)	Microfilming of Engineering Documents, 35mm, for Naval Ship Systems
MIL-M-38761/1(AS)	Microfilm and Tabulating Cards used for Recording Engineering Drawings and Associated Lists
MIL-M-38761/3(OS)	Microfilm and Microfilm Frame Deck used for Recording Engineering Drawings and Associated Data
MIL-M-38802 (USAF)	Microfilming, 16mm, and Reproductions of Technical Orders, Requirements for
MIL-P-63048 (TM)	Preparation of Equipment Publications Microfiche
MIL-M-81926 (AS)	Microfilm Data Retrieval Equipment System
MIL-M-81930 (AS)	Mil Spec. Manuals, Technical: 16mm Silver Halide Microfilm; Requirements for (See Note 1).
MIL-M-81931 (AS)	Mil Spec. Manuals, Technical: 16mm Duplicate Microfilm, Cartridge Loading and labeling, Packaging and Shipping; Requirements for. (See Note 1)
MIL-V-80240	Viewer Microfiche (24X and 48X) (See Note 2)
MIL-V-80241	Viewer/Printer Microfiche (24X and 48X) (See Note 2)

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Note 1 - These specifications, approved by Naval Air Systems Command, apply for use on the Navy Maintenance Information Automated Retrieval System (MIARS).

Note 2 - The Defense General Supply Center (DGSC), Defense Supply Agency (DSA), Richmond, Virginia buys and stocks, as items of supply, the following viewers and viewer/printers:

48x microfiche viewer - NSN 6730-00-074-2729
24x/48x microfiche viewer - NSN 6730-00-116-1618
24x/48x microfiche viewer/printer - NSN 6730-00-116-1620
48x microfiche viewer/printer - NSN 6730-00-044-3242

Whenever possible, this equipment should be obtained from DGSC-DSA citing the applicable National Stock Number (NSN).

Federal

G-C-116	Cards, Tabulating

STANDARDS

Federal

FED-STD-125	Film, Photographic and Film, Processed (for Permanent Record Use)
FED-STD-170	Film, Photographic, Black and White, Classification and Testing Methods

Military

MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of

HANDBOOKS

Military

MIL-HDBK-303	Micro-Reproduction of Engineering Documents
MIL-HDBK-331	Directory of DoD Engineering Data Repositories

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

Copies of these government documents may be obtained by authorized offices by completing DD Form 1425 and processing that form as indicated thereon.

6.3 Non-governmental documents

STANDARDS

American National Standards Institute

PH5.6-1968	100 Foot Reels for Processed 16mm and 35mm Microfilm, Dimensions

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PH2. 19-1959 Diffuse Transmission Density

PH4. 8-1971 Methylene Blue Method for Measuring Thiosulfate and Silver Densitometric Method for Measuring Residual Chemicals in Films, Plates, and Papers

(ANSI Standards may be obtained from the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018)

National Micrographics Association (NMA)

MS100-1971 Glossary of Micrographics (5th Edition)

(NMA standards may be obtained from the National Micrographics Association, 8728 Colesville Road, Silver Spring, MD 20910). See Note.

Note - Specifications and standards of technical societies and technical associations are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies. DoD adopted industrial standards may also be obtained by DoD activities from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

Preparing Activity:
ARMY-PA

Custodians:
ARMY-PA
NAVY-SA
Air Force-07
DSA-DH

Project Number:
MISC-0A64

Reviewer Activities:

Army-AD, AG, AR, AV, EA, EL, WC
Navy-AS, EC, NM, OS, SH, MC, NP, OP-09B17
Air Force-13, 15, 16, 17
Defense Supply Agency - DP, ES, GS, CS, IS, DD, LS
MISC DoD Activities-NA, NS
Civil Agencies and Coordinating Activities-AEC
Civil Agency Standardization Activities-ACO, GPO
Non-Standardization Activity DIA

User Activities:

Army-CE, IH, ME, MD, MI, MT, MU, SC, SG, TM, GL, WV
Navy-MS, NC, NO, OM, PE, SA, TD, YD
Air Force-None
Defense Supply Centers-CT, DM
Misc DoD Activities-DC
Civil Agencies and Coordinating Activities-DCG
Civil Agency Standardization Activities-FPI, NIH

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